

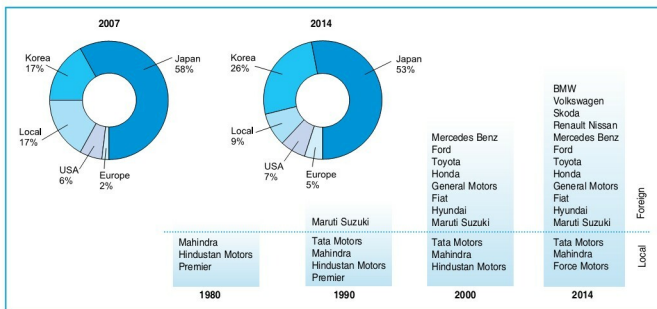


Fig. 2: Rapidly growing presence of global OEMs in India (Source: ACMA)

tive clusters across the country (Fig. 1). In comparison with the 1980s, when only a few local OEMs were present, India is now a manufacturing hub with significant presence of global and local OEMs (Fig. 2).

"From being a low-key supplier providing components exclusively to the domestic market, the industry has emerged as one of the key auto centres in Asia and is today seen as a significant player in the global automotive supply chain," says Nitin Pai, senior vice president, TATA Elxsi.

While specifying the opportunities that exist for automotive electronics, S. Ganesh Kumar, country sales manager, Atmel R&D India Pvt Ltd, opines, "Indian automotive product range varies from ultra-low-cost as the majority share up to mid-level cars. Each of these specific target products definitely requires some sort of electrification and electronic controls to enable highest efficiency and achieve some sort of comfort level. All these requirements create the potential for automotive electronics."

In addition, the two-wheeler segment, which dominates the automotive market in India, and commercial vehicle segment also show demand for electronics.

#### Adoption of automotive electron-

ics. In the early years of the automotive industry in India, manufacturers would use streamlined manufacturing processes as their competitive advantage as opposed to innovation. Electronics features, adopted by global car makers, would typically be given a miss by their Indian counterparts.

Process efficiency having reached its peak, the focus is now on innovation or additional novel features that are based primarily on electronic components. In fact, regional automotive manufacturers and design teams of global manufacturers present in India are increasing their focus on utilising technology to differentiate themselves from their peers.

The need for reduced production cost, coupled with increased quality along with customisable novel features, all within shorter business cycles, is driving an increased focus on the role of electronics within the region's automotive manufacturers.

Electronics content will become more advanced with each new generation of vehicles produced in India. "Implementation of properly-designed electronic systems can even reduce cost, while increasing flexibility and reliability of a vehicle," says Ramachandra. This suggests great

potential for the automotive electronics market.

**Consumer behaviour and demand.** Demand patterns of automotive electronics in the country are driven by local consumers and export requirements. Consumer needs typically focus around affordability, prestige, mobility, environmental conscience, safety, comfort and a host of other factors.

Affordability, for example, plays a huge role behind the miniscule share of the premium car segment in India. The ultra-low-cost (ULC) segment targeted by Tata Nano addresses the consumer's need for affordability, which is in head-on competition with the motorcycle—the current dominant form of private transportation in India. The major challenge for suppliers in this segment is to innovate technologically and business-wise in order to create a profitable and sustainable business model, keeping in mind price sensitivity.

A recent study, conducted by a team from *Electronics Bazaar*, among a select group of millennial or gen Y folks in the metro cities of India found that in-car electronics such anti-lock braking system (ABS), smart air-conditioning, power windows, and head and side airbags are